Weekly Homework 2

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March 12, 2026

Problem 1. When applying Newton's method to solve the equation $f(x) = 0$, we usually require that $f'(x_*) \neq 0$, i.e., the root x_* is a simple one. Without such a condition, Newton's method is still applicable to find x_* while the convergence is no longer quadratic. In the following let us assume that $f(x)$ is sufficiently smooth to avoid complications in theoretical analysis.

(a) Use Newton's method to solve $1 + \cos x = 0$ around $x_0 = 3$ and plot the convergence history.

(b) Let x_* be a root of $f(x)$ with multiplicity higher than one, i.e.,

$$f(x_*) = f'(x_*) = 0.$$

Show that Newton's method converges (locally) linearly around x_*.

(c) Let x_* be a root of $f(x)$ with multiplicity $m + 1$, i.e.,

$$f(x_*) = f'(x_*) = \cdots = f^{(m)}(x_*) = 0 \neq f^{(m+1)}(x_*).$$

We can modify Newton's method as

$$x_{k+1} = x_k - \frac{(m+1)f(x_k)}{f'(x_k)}$$

to achieve local quadratic convergence. Try to explain why such a modification improves the convergence.

(a) The convergence history is shown in the following figure.

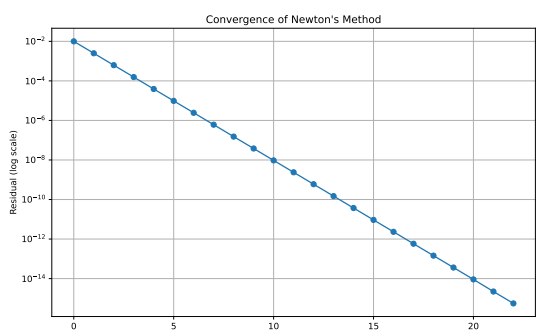

Figure 1: Convergence history of Newton's method for solving $1 + \cos x = 0$ around $x_0 = 3$.

(b) Notice that

$$\begin{aligned} f(x_k) - f(x_*) &= f'(x_*)(x_k - x_*) + \frac{1}{2} f''(x_*)(x_k - x_*)^2(1 + o(1)) \\ &= \frac{1}{2} f''(x_*)(x_k - x_*)^2(1 + o(1)), \\ f'(x_k) - f'(x_*) &= f''(x_*)(x_k - x_*)(1 + o(1)), \end{aligned}$$

Therefore, we have

$$\frac{f(x_k)}{f'(x_k)} = \frac{f(x_k) - f(x_*)}{f'(x_k) - f'(x_*)} = \frac{1}{2}(x_k - x_*)(1 + o(1)).$$

This implies that

$$|x_{k+1} - x_*| = \left| x_k - x_* - \frac{f(x_k)}{f'(x_k)} \right| = \frac{1}{2}|x_k - x_*|(1 + o(1)) \leq C|x_k - x_*|,$$

where $C > \frac{1}{2}$ is a constant. Hence, Newton's method converges linearly around x_*.

(c) Notice that

$$\begin{aligned} (m+1)(f(x_k) - f(x_*)) &= \frac{f^{(m+1)}(x_*)}{m!}(x_k - x_*)^{m+1}(1 + o(1)) \\ f'(x_k) - f'(x_*) &= \frac{f^{(m+1)}(x_*)}{m!}(x_k - x_*)^m(1 + o(1)), \end{aligned}$$

Therefore, we have

$$\frac{(m+1)f(x_k)}{f'(x_k)} = \frac{(m+1)(f(x_k) - f(x_*))}{f'(x_k) - f'(x_*)} = (x_k - x_*)(1 + o(1)).$$

This implies that

$$|x_{k+1} - x_*| = \left| x_k - x_* - \frac{(m+1)f(x_k)}{f'(x_k)} \right| = O(|x_k - x_*|^2),$$

which means that the modified Newton's method converges quadratically around x_*.

Problem 2. Show that Steffensen's method converges locally at least quadratically.

Proof. Let x_* be the root of $f(x)$, i.e., $f(x_*) = 0$. Let $g(x) = f(x + f(x))$. Since $f(x_*) = 0$, we have $g(x_*) = 0$ and $g'(x_*) = f'(x_*)(1 + f'(x_*))$. Therefore, if $f'(x_*) \neq 0$, we have

$$\begin{aligned} x_{k+1} - x_* &= x_k - x_* - \frac{(f(x_k))^2}{g(x_k) - f(x_k)} \\ &= x_k - x_* - \frac{(f(x_k) - f(x_*))^2}{(g(x_k) - g(x_*)) - (f(x_k) - f(x_*))} \\ &= x_k - x_* - \frac{(f'(x_*)(x_k - x_*) + f''(x_*)(x_k - x_*)^2/2)^2}{(f'(x_*))^2(x_k - x_*)}(1 + o(1)) \\ &= C|x_k - x_*|^2, \end{aligned}$$

where $C > 0$ is a constant. Hence, Steffensen's method converges locally at least quadratically. $\square$

Problem 3. In this exercise, you will determine the pH of rainwater by measuring the partial pressure of carbon dioxide (CO_2). For simplicity, we suppose that the only chemical reactions in rainwater are

$$\begin{aligned} H_2O &\rightleftharpoons H^+ + OH^-, \\ CO_2 + H_2O &\rightleftharpoons H^+ + HCO_3^- \\ HCO_3^- &\rightleftharpoons H^+ + CO_3^{2-}. \end{aligned}$$

The following nonlinear system of equations governs the chemistry of rainwater:

$$\begin{aligned} K_W &= [H^+][OH^-], \\ K_1 &= 10^6 \frac{[H^+][HCO_3^-]}{K_H \cdot p_{CO_2}}, \\ K_2 &= \frac{[H^+][CO_3^{2-}]}{[HCO_3^-]}, \\ [H^+] &= [OH^-] + [HCO_3^-] + 2[CO_3^{2-}], \end{aligned}$$

where $K_H = 10^{-1.46}$ is Henry's constant, and $K_1 = 10^{-6.3}$, $K_2 = 10^{-10.3}$ and $K_W = 10^{-14}$ are equilibrium constants.

Let us use $p_{CO_2} = 375$ (ppm), which was the partial pressure of CO_2 at MaunaLoa (Hawaii) in 2003. Estimate the corresponding pH of rainwater.

Solve this problem using the multivariate version of Newton's method and two variants of Broyden's method.

Solution. The solution to this problem is implemented in the file `Problem3.py`. The computed pH values, the corresponding residuals of the objective function at the solutions, and runtime are as follows:

```
Conditional Number: 7.0203943878352515

Newton's Method pH Solution: 5.592642901145539
Newton's Method Residual: 8.301806609254383e-18
Newton's Method Time: 0.0002560615539550781

Good Broyden's Method pH Solution: 5.592642901151882
Good Broyden's Method Residual: 8.292409394178671e-17
Good Broyden's Method Time: 0.0007925033569335938

Bad Broyden's Method pH Solution: 5.592642901145041
Bad Broyden's Method Residual: 2.4390146082866566e-18
Bad Broyden's Method Time: 0.000942230224609375
```

Figure 2: Computed pH values, residuals of the objective function at the solutions, and runtime for Newton's method, Good Broyden's method, and Bad Broyden's method.

□

Problem 4. Let $p(x)$ be a polynomial of degree n with roots $\mu_1, \mu_2, \ldots, \mu_n$. Suppose that $\gcd(p(x), p'(x)) = 1$, i.e., $p(x)$ has no repeated roots. Show that

$$\sum_{i=1}^{n} \frac{1}{p'(\mu_i)} = 0.$$

Proof. Without loss of generality, we can assume that $p(x) = \prod_{i=1}^{n}(x - \mu_i)$. Then,

$$p'(\mu_i) = \prod_{j \neq i}(\mu_i - \mu_j).$$

Therefore, we have

$$\sum_{i=1}^{n} \frac{1}{p'(\mu_i)} = \sum_{i=1}^{n} \prod_{j \neq i} \frac{1}{\mu_i - \mu_j}$$

Notice that the term $\prod_{j \neq i}(\mu_i - \mu_j)^{-1}$ is exactly the coefficient of x^{n-1} in the Lagrange basis polynomial

$$L_i(x) = \prod_{j \neq i} \frac{x - \mu_j}{\mu_i - \mu_j}.$$

Consider the interpolation of the constant function $f(x) \equiv 1$ at points $\mu_1, \ldots, \mu_n$. Since the degree of $f(x)$ is 0, which is less than n (assuming $n \geq 2$), the Lagrange interpolation formula gives exactly $f(x)$:

$$1 = \sum_{i=1}^{n} f(\mu_i) L_i(x) = \sum_{i=1}^{n} L_i(x).$$

Comparing the coefficient of x^{n-1} on both sides:

- On the LHS, the coefficient of x^{n-1} is 0 (since $n \geq 2$).
- On the RHS, the coefficient is $\sum_{i=1}^{n} \frac{1}{\prod_{j \neq i}(\mu_i - \mu_j)}$.

Thus, we conclude that

$$\sum_{i=1}^{n} \frac{1}{p'(\mu_i)} = 0.$$

□

Problem 5. Polynomial interpolation provides one way to approximate a given function. For instance, let $-\pi = x_1 < x_2 < \cdots < x_n = \pi$ be equally spaced interpolation nodes. The interpolating polynomial that passes through all the $(x_i, \sin x_i)$'s can be used to approximate the sine function $f(x) = \sin x$. Try to visualize the difference between the interpolating polynomial and the sine function for a few different choices of n.

Do the same for $f(x) = (1 + 25x^2)^{-1}$ over $[-1, 1]$.

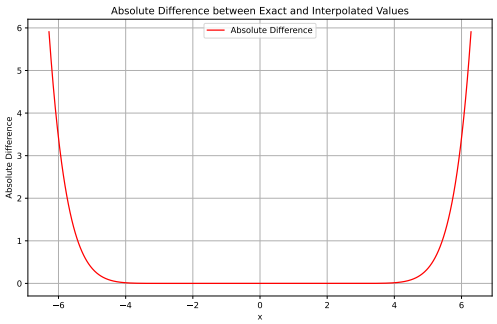

Figure 3: Absolute difference of interpolation of $f(x) = \sin x$ using 10 equally spaced nodes.

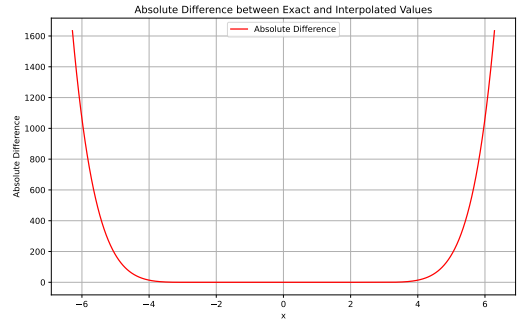

Figure 4: Absolute difference of interpolation of $f(x) = (1 + 25x^2)^{-1}$ using 10 equally spaced nodes.

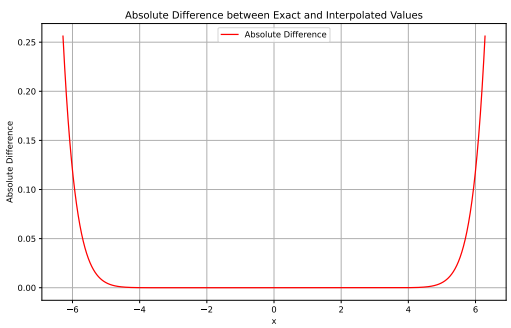

Figure 5: Absolute difference of interpolation of $f(x) = \sin x$ using 15 equally spaced nodes.

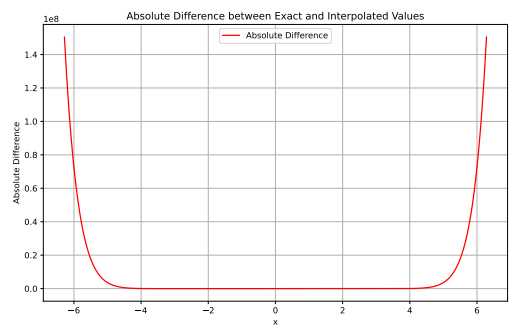

Figure 6: Absolute difference of interpolation of $f(x) = (1 + 25x^2)^{-1}$ using 15 equally spaced nodes.

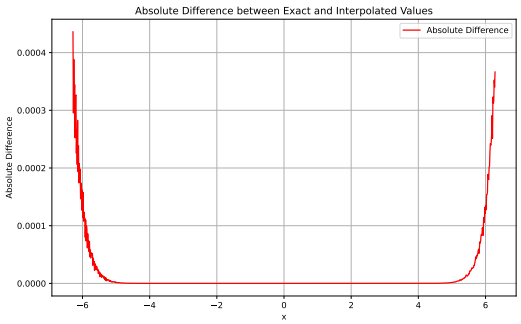

Figure 7: Absolute difference of interpolation of $f(x) = \sin x$ using 20 equally spaced nodes.

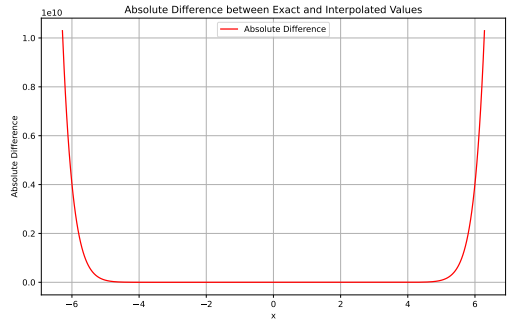

Figure 8: Absolute difference of interpolation of $f(x) = (1 + 25x^2)^{-1}$ using 20 equally spaced nodes.